

Jacob Mattie

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Summary

Full-Stack Engineer with a mathematics background. Specialized in data pipelines, databases, and automation.

Python, C++, SQL, Bash | Linux, Windows | Git, AWS, Django

Work Experience

Investment Analyst / Full Stack Engineer

Aligned Capital Partners

Vancouver, BC

11/2024 - Ongoing

- Built ETL **data pipelines** merging over 15 sources (files, APIs, web scraping) into a PostgreSQL schema, enabling the use of **machine learning** techniques for investment decisions.
- Configured a **Linux-based server stack** (Fedora, Gunicorn, Nginx) to host a **Django** application built from scratch, making data analytics and automation tools accessible to administrative staff.
- Automated recurring tasks in Python, reducing office-wide labour load by 15% and reducing the likelihood of human error.

Test Technologist

IBC Technologies

Burnaby, BC

6/2022 – 9/2023

- Automated testing with IoT microcontrollers and PLC devices, creating **cloud-connected data pipelines** to send controls and receive data from validation units.
- **Automated data analysis** of high-volume datasets using MATLAB, reducing processing time from 20 minutes to under 3 minutes per iteration.

IT Technician (Internship)

BC Cancer Research Centre

Vancouver, BC

09/2018 – 01/2019

- Diagnosed and resolved technical issues across Windows, Mac, and Linux environments in a large-scale research facility, solidifying **adaptable problem-solving** skills.
 - Configured and maintained network infrastructure and digital policy to ensure **cybersecurity** compliance and reliable data transmission across departments.
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Education

Simon Fraser University

B.A.Sc in Mathematics

01/2019 - 05/2022

Volunteering & Leadership

Triumf - ASPIRE Astrochemistry Lab

Automation Technologist

Vancouver, British Columbia

05/2025 - Ongoing

- Contributed to the publication of 2 papers:
 - Astrochemistry; In Preparation
 - Control Systems; In Preparation
- Developed a Bash-based installer for EPICS, an industry-standard **lab control framework** for physics laboratories, mitigating a significant entry labour barrier for new students and staff.
- Integrated a Python motor controller into the laboratory control system, **eliminating effective experimental variance** and reducing need for active engagement with the experiment over its runtime.